BACKUP POWER PRO BY KEY ELECTRIC

Generator Power Guide

Know what your portable generator can actually run, how to connect it safely, and what to look for before the next outage.

📄 8 pages ⚡ Wattage chart included 📋 Printable worksheet

FREE GUIDE

Serving Pickens, Greenville & Spartanburg, SC

★★★★★
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SECTION 01

Generator Safety

Portable generators save the day during outages, but they demand respect. Follow these rules every single time.

1. **Never run indoors**
   Generators produce carbon monoxide — an odorless, deadly gas. Always run your generator outside, at least 20 feet from any window, door, or vent.

2. **Use heavy-duty cords**
   If running extension cords, use outdoor-rated, grounded cords (12-gauge minimum). Never daisy-chain multiple cords together.

3. **Keep it dry**
   Operate on a dry surface under an open canopy. Never touch a generator with wet hands. Water and electricity don't mix.

4. **Let it cool before refueling**
   Shut the generator off and wait at least 5 minutes before adding fuel. Gasoline on a hot engine can ignite instantly.

5. **Don't back-feed your panel**
   Plugging a generator directly into a wall outlet is illegal and deadly — it sends power back through the grid and can electrocute utility workers.

6. **Install a proper inlet**
   A code-compliant inlet box and interlock kit is the only safe way to connect a portable generator to your home's electrical panel.

⚠️ **CO kills fast.**
Install battery-operated CO detectors on every level of your home. If you feel dizzy, nauseous, or get a headache during a power outage — get outside immediately.

SECTION 02

How Wattage Works

Every generator has two wattage ratings. Understanding both is the key to knowing what you can run without overloading.

Running Watts

The continuous power your generator can sustain. This is your real budget — plan around this number.

Starting Watts

The brief surge when a motor kicks on (fridges, ACs, pumps). Usually 2–3x running watts, lasts a few seconds.

THE RULE

Add up running watts of everything you want on.

Stay under your generator's running watt rating. Leave 10–20% headroom for safety.

Quick Example

Refrigerator	150W
5 LED lights	50W
WiFi router	20W
Phone chargers (×3)	30W
Laptop	60W
Total	**310W**

A 3,500W generator handles this easily with room for a fan or TV.

SECTION 03

Appliance Wattage Chart

Average wattage for common household appliances. Check the label on yours for exact specs.

APPLIANCE	RUNNING (W)	STARTING (W)	PRIORITY
Refrigerator / Freezer	150–200	800–1,200	Essential
LED Lights (per bulb)	8–15	—	Essential
WiFi Router / Modem	15–25	—	Essential
Phone Charger	5–10	—	Essential
Laptop	50–80	—	Essential
Sump Pump (⅓ HP)	800	1,200–1,800	Essential
Well Pump (½ HP)	1,000	1,800–2,500	Essential
Window A/C (small)	500–700	1,000–1,500	Comfort
Central A/C (3-ton)	3,000–3,500	4,500–5,000	High draw
Space Heater	1,000–1,500	—	Comfort
Microwave	800–1,200	—	Comfort
Coffee Maker	600–900	—	Comfort
TV (LED, 50")	80–120	—	Comfort
Ceiling Fan	50–80	—	Comfort
Garage Door Opener	400–600	800–1,200	Useful
Washing Machine	500–700	1,200–1,800	Can wait
Electric Dryer	4,000–5,000	5,000–6,000	High draw
Electric Water Heater	4,000–4,500	—	High draw
Hair Dryer	1,200–1,800	—	Can wait
Toaster	800–1,200	—	Can wait

SECTION 04

What Can Your Generator Run?

Here's what to realistically expect at each common generator size. Plan around running watts, not peak.

2,000W Running
Honda EU2200i, similar inverters

- Fridge + a few lights + phone chargers + router
- One thing at a time — no stacking big loads
- Perfect for the basics during a short outage

3,500W Running
Predator 3500, Champion 3500

- Everything above + TV + fan + laptop
- Can handle a small window A/C if you're careful
- The sweet spot for most homes

5,000W+ Running
Predator 4375, Champion 5000+

- Everything above + sump pump + well pump
- Multiple rooms with lights and outlets
- Can handle bigger loads — but still no central A/C or dryer

7,500W+ Running
Large portable generators

- Nearly everything except central A/C and electric dryer
- Well pump + fridge + lights + multiple appliances
- Very capable — but heavy, loud, and burns more fuel

SECTION 05

Extension Cords vs. A Proper Inlet

Most generator owners run cords through a window. It works — but here's what you're risking.

Extension Cords

× Open windows let in rain, cold, CO
× Cords overheat → fire hazard
× Only powers what's plugged in
× Violates electrical code
× Can void insurance during a claim

Inlet + Interlock Kit

✓ All windows stay closed
✓ Power runs through your panel
✓ Choose which circuits to power
✓ Fully code-compliant with permit
✓ Insurance-friendly, protects lineworkers

How It Works

An inlet is a weatherproof plug mounted on the outside of your home. Your generator connects to it with a single heavy-duty cord. Inside, an interlock kit on your breaker panel lets you safely switch from utility power to generator power. It's a permanent, one-time installation.

The Process

1. **Video Assessment · 10 min**
   Point your phone at your breaker panel. We check the brand, size, and layout. No in-home visit needed.

2. **Quote · Same Day**
   Your exact price, no surprises. Most installs fall between $1,400 and $2,000 depending on panel type.

3. **Installation · One Day**
   We pull the permit, mount the inlet, wire it to your panel, install the interlock kit. Your power stays on the entire time.

4. **Inspection**
   A county inspector verifies the work. You get documentation that everything is code-compliant.

YOUR PLAN

Build Your Power Budget

Write in the appliances you want to power and add up the running watts.
Stay under your generator's running watt rating.

APPLIANCE	RUNNING WATTS
1. _____	
2. _____	
3. _____	
4. _____	
5. _____	
6. _____	
7. _____	
8. _____	
9. _____	
10. _____	
Total Running Watts	_____ W
My Generator's Running Watts	_____ W

Want to skip the extension cords?

We install a code-compliant inlet so your generator powers your home through your panel. One day. $1,400 – $2,000.

Book Your Free Video Assessment backuppowerpro.com

BACKUP POWER PRO

by Key Electric

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